

Managing Processes

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- Processes are nothing but the running file
- Memory - (processes depend upon memory)
- Storage - (data storage)
- Processes - (Denote the speeds of process)
- Types of process
 - Shell process / job
 - Daemons process / system process (by default system run the process. Run in background)

Command

1. top

This is the output of the top command in Linux, providing real-time information about system performance, resource usage, and running processes. Let's break it down:

1. Top Line: System Information

top - 07:35:52 up 1:01, 1 user, load average: 0.00, 0.00, 0.00

- **top**: The command you executed.
- **07:35:52**: The current system time.
- **up 1:01**: How long the system has been running (1 hour and 1 minute).
- **1 user**: Number of users currently logged into the system.
- **load average: 0.00, 0.00, 0.00**: System load averages over the last 1, 5, and 15 minutes. These values are very low, indicating the system is almost idle (0.00 means no load).

2. Tasks: Process Overview

Tasks: 104 total, 1 running, 103 sleeping, 0 stopped, 0 zombie

- **Tasks: 104 total**: Total number of processes running or in the process table.
- **1 running**: Number of processes actively running.
- **103 sleeping**: Number of processes that are idle/waiting for an event.
- **0 stopped**: Processes that are stopped (paused).
- **0 zombie**: Number of zombie processes (none here).

3. CPU Usage:

%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni, 90.9 id, 0.0 wa, 0.0 hi, 0.0 si, 9.1 st

- **0.0 us**: Percentage of CPU used by user processes.
- **0.0 sy**: Percentage of CPU used by system processes.
- **0.0 ni**: Percentage of CPU used by processes with a nice (priority) value.
- **90.9 id**: Percentage of CPU that is idle (not in use).
- **0.0 wa**: Percentage of CPU waiting for I/O operations.
- **0.0 hi**: Percentage of CPU used to handle hardware interrupts.
- **0.0 si**: Percentage of CPU used to handle software interrupts.
- **9.1 st**: Percentage of CPU stolen by hypervisors (likely you're on a virtualized environment, such as AWS).

4. Memory Usage:

MiB Mem : 957.4 total, 467.4 free, 320.7 used, 321.2 buff/cache

- **957.4 total**: Total physical memory available.
- **467.4 free**: Memory currently not in use.
- **320.7 used**: Memory actively used by processes.
- **321.2 buff/cache**: Memory used by buffers and cache (to speed up operations).

5. Swap Usage:

MiB Swap: 0.0 total, 0.0 free, 0.0 used. 636.8 avail Mem

- **0.0 total:** Total swap space (none is configured).
- **0.0 free:** Swap space available (none, because swap isn't configured).
- **0.0 used:** Swap space in use (none).
- **636.8 avail Mem:** Approximate memory available for starting new applications without swapping.

Key Observations:

1. **Load:** The system is very lightly loaded (load averages are 0.00).
 2. **CPU Usage:** Most of the CPU (90.9%) is idle, and 9.1% is stolen, indicating it's on a virtualized environment like AWS.
 3. **Memory:** Memory usage is moderate; a significant portion (321.2 MiB) is used for buffers/cache.
 4. **Swap:** No swap space is configured, which is common for lightweight AWS instances.
- Top command are used to see the system background process. The root user manage the process. live date are to be show in the top command...

In the Linux top command, the columns represent various metrics for processes running on the system. Here's what each column means:

Column Descriptions:

1. **PID:**
 - **Process ID:** A unique identifier for each process.
2. **USER:**
 - **User Name:** The name of the user who owns the process.
3. **PR:**
 - **Priority:** The priority level of the process. Lower numbers indicate higher priority.
4. **NI:**
 - **Nice Value:** The "niceness" value of the process. This affects its priority. Lower values give higher priority (can range from -20 to 19).
5. **VIRT:**
 - **Virtual Memory Size:** The total amount of virtual memory used by the process, including code, data, and shared libraries, plus memory that has been swapped out.
6. **RES:**
 - **Resident Memory Size:** The portion of the process's memory that is currently in physical RAM.
7. **SHR:**
 - **Shared Memory:** The amount of memory shared with other processes.
8. **S:**
 - **State:** The current state of the process:
 - **R:** Running
 - **S:** Sleeping
 - **D:** Uninterruptible sleep (usually I/O)
 - **Z:** Zombie (terminated but not reaped by its parent)
 - **T:** Stopped (by a signal or user)
9. **%CPU:**
 - **CPU Usage:** The percentage of CPU time used by the process.
10. **%MEM:**
 - **Memory Usage:** The percentage of physical RAM used by the process.
11. **TIME+:**
 - **CPU Time:** The total CPU time the process has used since it started (in the format of minutes:seconds.tenths).
12. **COMMAND:**
 - **Command Name:** The name or command line of the process.

This information helps system administrators monitor and manage system performance, resource usage, and troubleshoot issues.

2. sleep

- Command sleep 10 (sec)
- They are not used in terminal

- CTRL + z = stopped the command restart the process
 - CTRL + c = cancel the command using
3. sleep 200 &
 - The command are process in background using the & for any command least but the terminal are to be used
 4. fg %2
 - To restate the stopped process 2 are denote the job ID they are provide the system
 5. bg %5
 - To restate the process in background and used the terminal
 6. jobs
 - They are to show the jobs of command to provide the system
 7. ps -e
 - standard syntax
 8. ps -aux
 - (system using BSD syntax) They are to be show the background running process
 9. pkill
 - Only running process are to be remove
 10. killall
 - Running and stop process are to be remove
 11. ps -elf
 - The ps -elf command in Linux is used to display a detailed (long) listing of all processes running on the system
 - -e: Show all processes on the system, not limited to the current user or session.
 - -l: Use a long listing format to display detailed information about each process.
 - -f: Display full-format listing (shows additional information such as the parent process ID and command arguments).

Columns in ps -elf Output

Here's what each column in the output means:

Column	Description
F	Flags associated with the process.
S	Process state (e.g., R = Running, S = Sleeping, Z = Zombie, T = Stopped).
UID	User ID of the process owner.
PID	Process ID (unique identifier for each process).
PPID	Parent Process ID (ID of the process's parent).
C	CPU usage percentage used recently.
PRI	Priority of the process (lower value = higher priority).
NI	Niceness value (priority adjustment, -20 to 19).
ADDR	Memory address of the process (kernel space).
SZ	Size of the process in memory (in pages).
WCHAN	If sleeping, the kernel function in which the process is waiting.

TIME	Total CPU time consumed by the process.
CMD	The command that started the process.

Sample Output of ps -elf

plaintext

Copy code

```
F S UID PID PPID C PRI NI ADDR SZ WCHAN TIME CMD
4 S 0 123 1 0 20 0 - 500 wait 00:00:01 /usr/sbin/apache2
4 R 0 456 123 1 20 0 - 100 - 00:00:02 /usr/bin/python3 script.py
0 S 1000 789 456 0 20 5 - 200 futex 00:00:00 /usr/bin/vim file.txt
```

Explanation of the Sample

1. First Process:

- PID 123 is a sleeping process (S), owned by UID 0 (root).
- It is an Apache process (/usr/sbin/apache2).
- It has used 1 second of CPU time.

2. Second Process:

- PID 456 is a running process (R) under UID 0.
- It is running a Python script.

3. Third Process:

- PID 789 is a sleeping process (S) owned by UID 1000 (a user).
- It is running vim.

12. ps

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